AMD DEVELOPER HACKATHON: ACT III · INTELLIGENT INDUSTRY

worksight.

Few-shot visual inspection for quality engineers

Onboard a part from 10–20 good photos. Get a heatmap, the closest good reference and an explanation from **Gemma on AMD**. The engineer makes the call.

Live: **web-production-d9743.up.railway.app**
Code: **github.com/DesertEaglesTech/worksight** · MIT



Real test image: VisA pcb1, bent pins (Amazon, CC BY 4.0)

THE PROBLEM

A new part on the line needs inspection today, not after a defect dataset exists

WHO

The quality engineer

Owns the pass / fail decision for a part and camera view. Has plenty of photos of good parts and very few of defects.

TODAY

Supervised models wait for defects

A typical defect classifier needs many labelled defect images and a training run for every new part. Defects are rare, so the data arrives last.

AND

A score alone is not evidence

The engineer has to see **where** the part differs, **compared with what**, and record their own decision.

THE PRODUCT

Onboard, inspect, explain, decide

1 · ONBOARD

~9 s

Profile from 10–20 photos of good parts. No defect labels, no training run. Measured: 20 references on Railway CPU.

→

2 · INSPECT

0.1–0.3 s

Anomaly score, heatmap of unusual regions and the most similar good reference. Normal / review / anomaly.

→

3 · EXPLAIN

Gemma

Gemma-4-31B-IT on AMD Token Factory describes what looks wrong and what to check, using the part's documentation.

→

4 · DECIDE

Human

The engineer records their decision and note. It is stored separately; the model result is never overwritten.

Live check on the hosted app: VisA pcb1 “bent pins” → **Fail**, score 0.279 against threshold 0.094, heatmap on the pins, Gemma description returned.

APPLICATION OF TECHNOLOGY

Gemma on AMD explains every inspection

Browser · Next.js

Operator workspace. A server-side proxy keeps the API key out of the browser.

FastAPI · PyTorch detector

Frozen ResNet-18 patch memory, nearest-patch retrieval, heatmap. ROCm-aware device selection.

AMD Token Factory · Gemma-4-31B-IT

Gets the photo, heatmap overlay, closest good reference, detector decision and part documentation. Fallback: DeepSeek-V4.1-Flash, also on Token Factory.

- **Grounded:** Gemma reads the same evidence the engineer sees, plus the part's own PDF or text documentation.
- **Bounded:** the detector's decision and score are passed in and never changed. The report is labelled assistive.
- **Traceable:** each report records which model answered and its response time.
- **Resilient:** two retries on server errors, then the fallback model.

Next Detector inference on an AMD GPU: **scripts/amd_run.sh** runs the smoke test and benchmark with --require-amd. Not run yet; hosted detector is CPU.

MEASURED RESULTS

Real boards, held-out test, no test-set tuning

VISA	AUROC	SCORE ≥ THRESHOLD		SENT TO ENGINEER	
		RECALL	FALSE ALARMS	DEFECTS	GOOD BOARDS
pcb1	0.905	44%	5%	**56%**	7%
pcb2	0.884	59%	6%	**77%**	13%
pcb3	0.875	46%	4%	**58%**	10%
pcb4	0.992	87%	2%	**96%**	4%

“Sent to engineer” = anomaly or review (±10% band around the threshold, fixed before the test). 20 references from the official train normals; full official test split (~100 normal + 100 defect images per board). Design chosen on 8 other VisA categories. CPU, p50 194–304 ms; not an AMD run.

LIVE SPOT CHECK

10-image pcb1 batch on the hosted app: **all 5 defects flagged**, one good board sent to **review**.

HONEST LIMITS

Above the threshold alone, recall is ~45% on pcb1/pcb3; with the review band 56–58% of defects reach a person. The score is a distance, not a probability: Worksight is a triage tool, not an autonomous rejecter.

Source: docs/BENCHMARK.md, docs/STATUS.md

HUMAN + AI WORKFLOW

The engineer stays in charge of every change

FEEDBACK

Engineer confirms or rejects the model result. Saved feedback is immutable.

→

CANDIDATE

Confirmed-normal parts build a new profile version. It is not active yet.

→

GATE

Compared with the current version on a frozen, labelled validation set. Any new false positive or false negative blocks it.

→

ACTIVATE

Explicit activation, audit record, rollback to an earlier approved version.

Old inspections keep the version they used, so every past decision can be traced to the exact references behind it.

MARKET AND COMPETITION

A growing market built on labelled defects

MARKET

+$26.8B

Forecast growth of the AI visual inspection system market, 2024–2029, at a 38.8% CAGR, driven by quality control in manufacturing (Technavio, 2025).

APPROACH	STARTS FROM	FOR THE OPERATOR	PRICING
Enterprise suites e.g. Cognex Deep Learning	Labelled training images	Vendor tooling	Quote: hardware + licence per line
No-code platforms e.g. LandingLens	Label and train in the browser	Hosted app	Free tier + enterprise
Open-source libraries e.g. anomalib	Good images	None: code for ML engineers	Free
Worksight	**10–20 good photos, ~9 s**	**Heatmap, Gemma explanation, engineer decision, gated versions**	Per camera view (planned)

Sources: Technavio, AI Visual Inspection System Market 2025–2029; vendor listings on Capterra and Dupple (2026)

BUSINESS VALUE AND PLAN

Inspection from day one of a new part

- **No defect dataset to start:** a part is onboarded from good photos the line already has, in seconds instead of a labelling and training cycle.
- **Engineer time on the right parts:** normal / review / anomaly; 56–96% of defects reach a person while 4–13% of good boards do (VisA PCB).
- **Evidence for every decision:** heatmap, closest reference, Gemma explanation and the engineer's note, stored together.
- **Who pays:** the quality team of a line. Planned model: subscription per camera view, or on-premises on AMD hardware for plants that keep images on site.
- **Pilot:** one line, one camera; measure engineer time per part and escaped defects against today's inspection.

BUILT

Live web app, FastAPI backend, Gemma reports on AMD, versions with gate and rollback. 36 backend + 39 web tests.

NEXT

1. Detector on AMD Instinct via ROCm (script ready)
2. Batch and stats screens (API ready)
3. Edge station on AMD Ryzen AI
4. First pilot line